# Statistical Analysis of Gender Pay Disparities in University Faculty Salaries

DATA 557 W26 Final Project - Deliverable 1

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## 1. Introduction

Gender pay disparities have been widely documented across many labor markets, including academia. Because salary discrimination based on sex is illegal in the United States, differences in salary between male and female faculty members are often debated. Some argue that disparities may arise from discrimination, while others attribute them to differences in experience, academic field, promotion history, or administrative responsibilities. These factors may also be interrelated, making it challenging to isolate the independent effect of gender on salary disparities.

This project investigates whether gender-based salary disparities exist among faculty at a particular U.S. university. We examine whether systematic differences in salary or promotion outcomes exist between male and female faculty members.

Specifically, we address four questions:

1. Does wage discrimination exist in the starting salaries of faculty members (i.e., salaries in the year hired)?
2. Does wage discrimination exist in granting promotions from Associate Professor to Full Professor?
3. Overall: Is there wage discrimination in salaries at the university?
4. What issues are involved in generalizing our results?

## 2. Data

The dataset contains salary records for faculty members employed at a single U.S. university between 1976 and 1995. Each observation corresponds to a faculty member in a particular year. The dataset includes 19,788 salary records representing 1,597 unique faculty members.

Key variables include:

- sex - gender of the faculty member (Male or Female)
- salary - monthly salary in dollars
- rank - faculty rank (Assistant, Associate, Full)
- field - academic field category (Arts, Professional, Other)
- yrdeg - year highest degree was attained
- startyr - year faculty member was hired
- year - observation year
- admin - indicator for administrative duties

The derived variable, experience, was created during preprocessing. It is calculated as the difference between the observation year and the year hired.

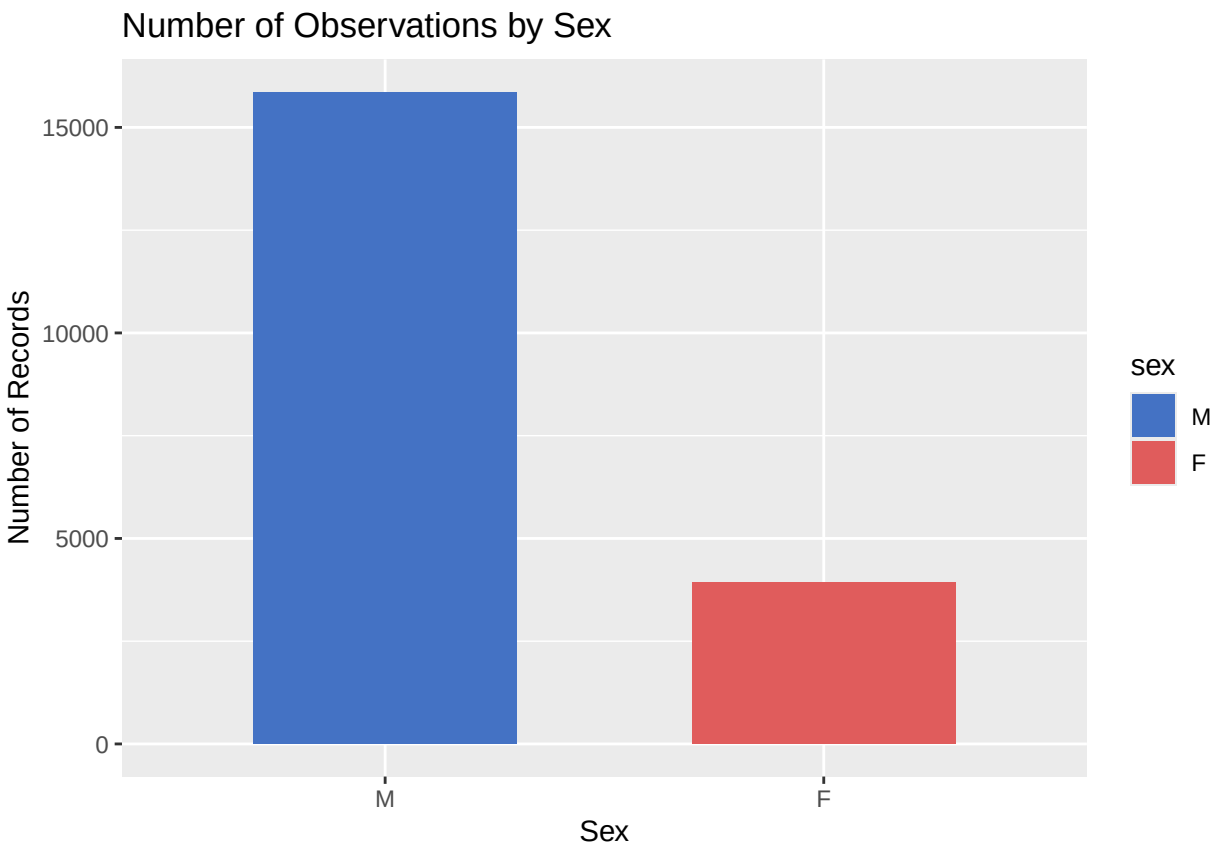
Observations with invalid values, such as missing rank information, were removed during data cleaning.

## 3. Exploratory Data Analysis

We conduct a preliminary exploratory data analysis (EDA) to examine the structure of the dataset and identify general patterns in salaries across gender, rank, and time.

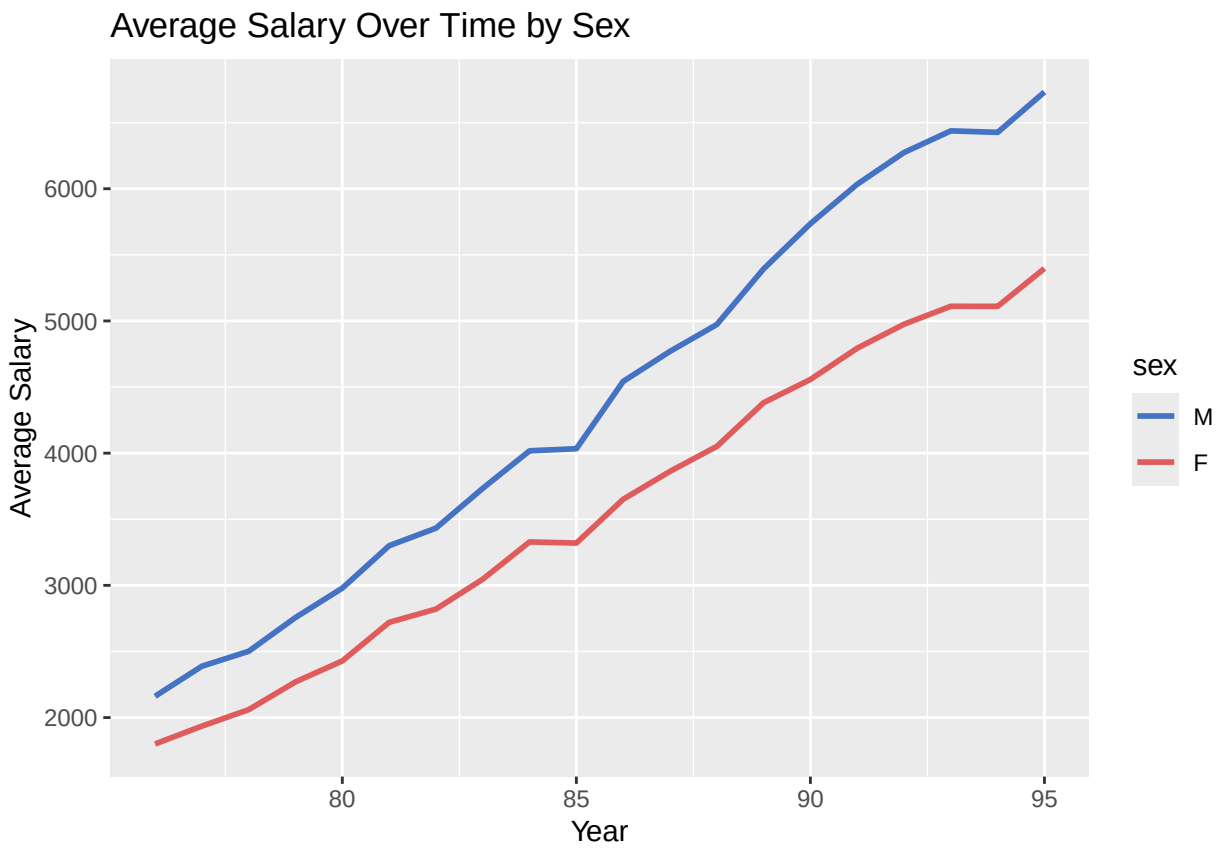
### Gender Composition of the Dataset

We begin by examining the distribution of observations by gender. The dataset contains substantially more observations for men than for women. Although this imbalance does not invalidate statistical comparisons, it is important to recognize that differences in sample size may affect the variability of estimates and the precision of statistical tests.



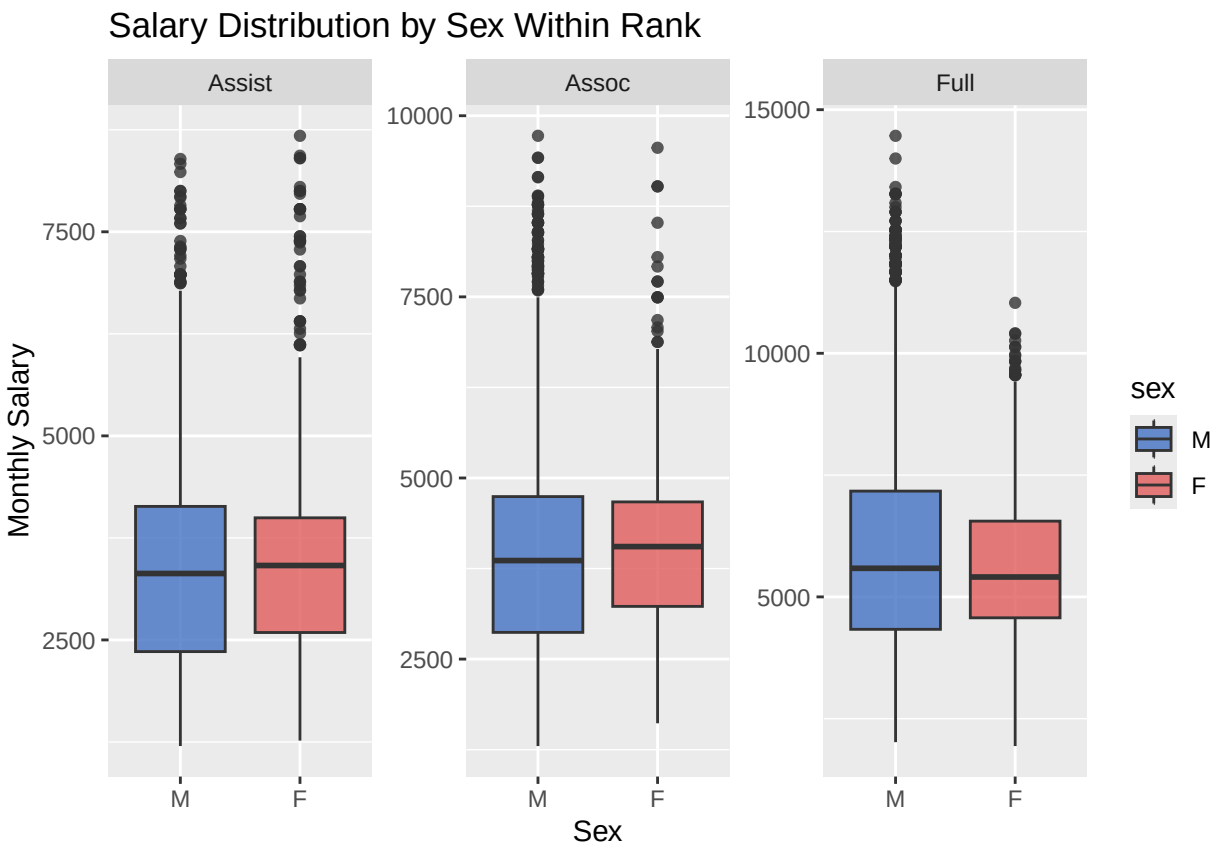

### Salary Trends Over Time by Sex

The time series plot shows that salaries increase steadily between 1976 and 1995 for both groups. Across all years, the average salary for men appears higher than that for women. At first galance, this pattern may suggest the presence of a gender wage gap. However, because academic salaries are strongly structured by rank, differences in the distribution of men and women across ranks could influence these aggregate averages. This possibility motivates a closer examination of salary distributions within each rank.




**Salary Distribution by Sex Within Rank**

The boxplots show that salaries increase substantially with rank, with Full Professors earning the highest salaries on average, followed by Associate Professors and then Assistant Professors. Within each rank, male and female salary distributions appear broadly similar, with the median female salary slightly higher in both Assistant and Associate groups. These patterns suggest that some observed salary differences may arise from differences in rank composition as well as potential within-rank wage gaps. This suggests an important statistical issue known as Simpson's paradox, where patterns observed in aggregate data can differ from patterns observed within subgroups. If men are disproportionately represented in higher-paying ranks, the overall average salary for men may appear higher even if salaries are similar within individual ranks.



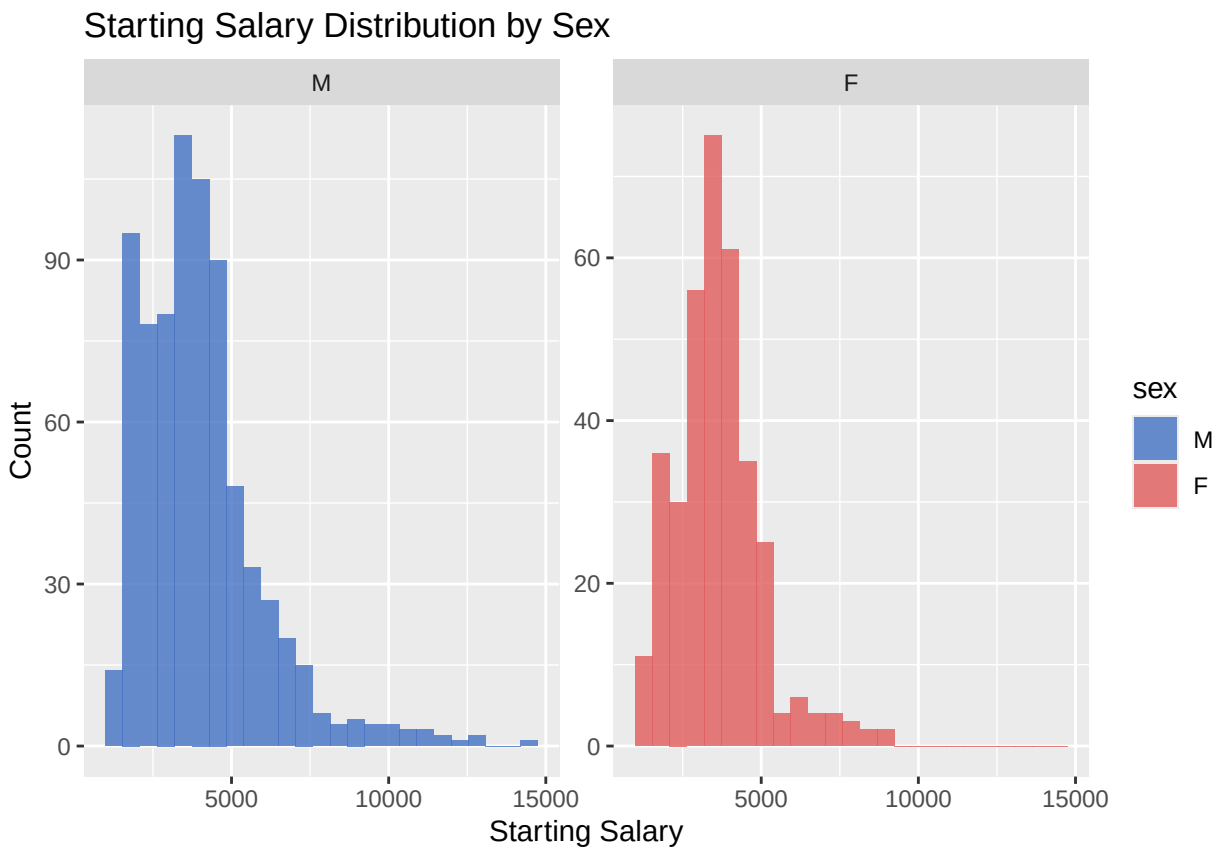

These exploratory plots emphasize the importance of accounting for rank when analyzing gender differences in salaries, which we address explicitly in our Question 2 analysis section.

## 4. Q1: Starting Salary Discrimination

To investigate whether gender discrimination exists in starting salaries, the dataset was filtered to include only the first year of employment for each faculty member (where year = startyr). This resulted in 1,107 starting salary observations. The mean starting salaries were $4,015.55 for men and $3600.32 for women. The mean starting salary for male faculty was approximately $415 higher than that of female faculty. Although the majority of newly hired faculty were Assistant Professors, some individuals were hired directly at the Associate or Full Professor ranks.

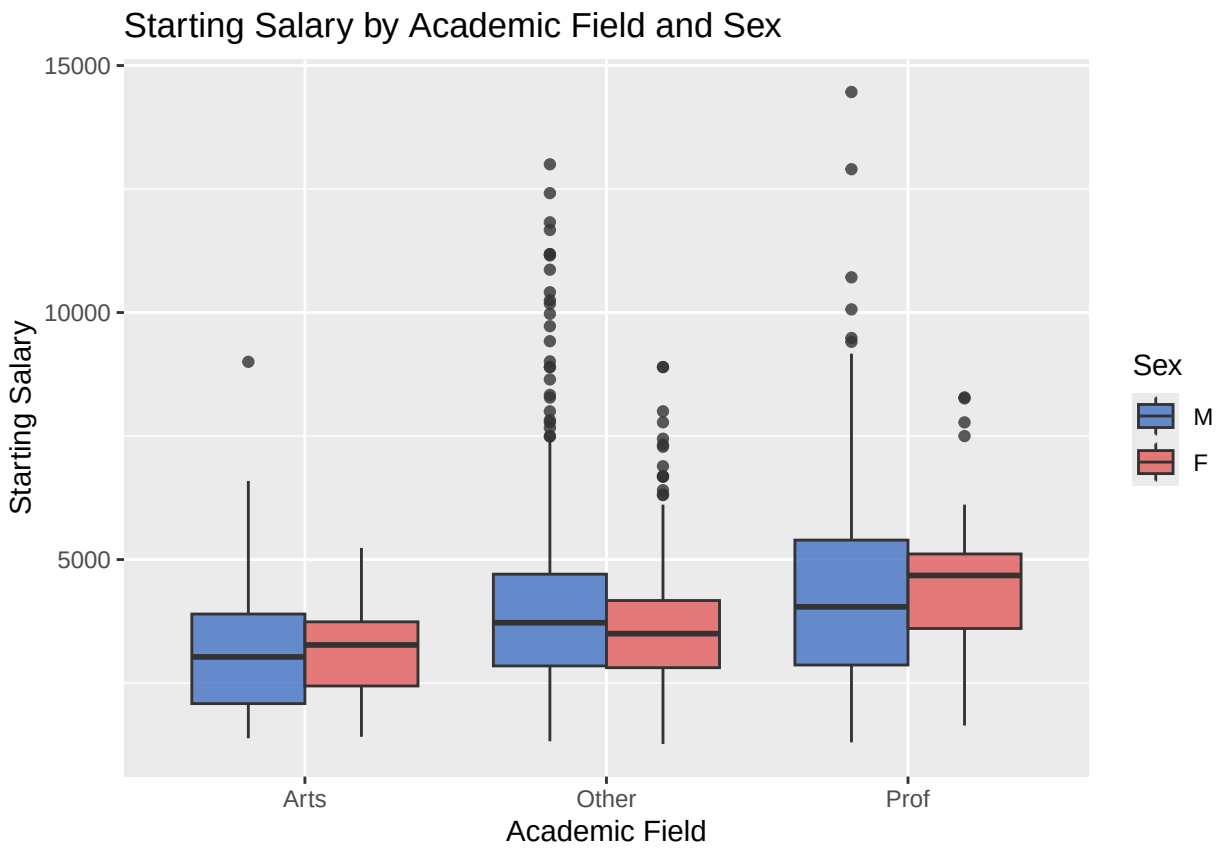
### Starting Salary Distribution by Sex

We first examine the distribution of starting salaries separately for male and female faculty members. The histograms indicate that starting salaries for both groups follow a right-skewed distribution. While the overall shapes of the distributions are similar, the distribution for men shows a greater number of observations above $5000.



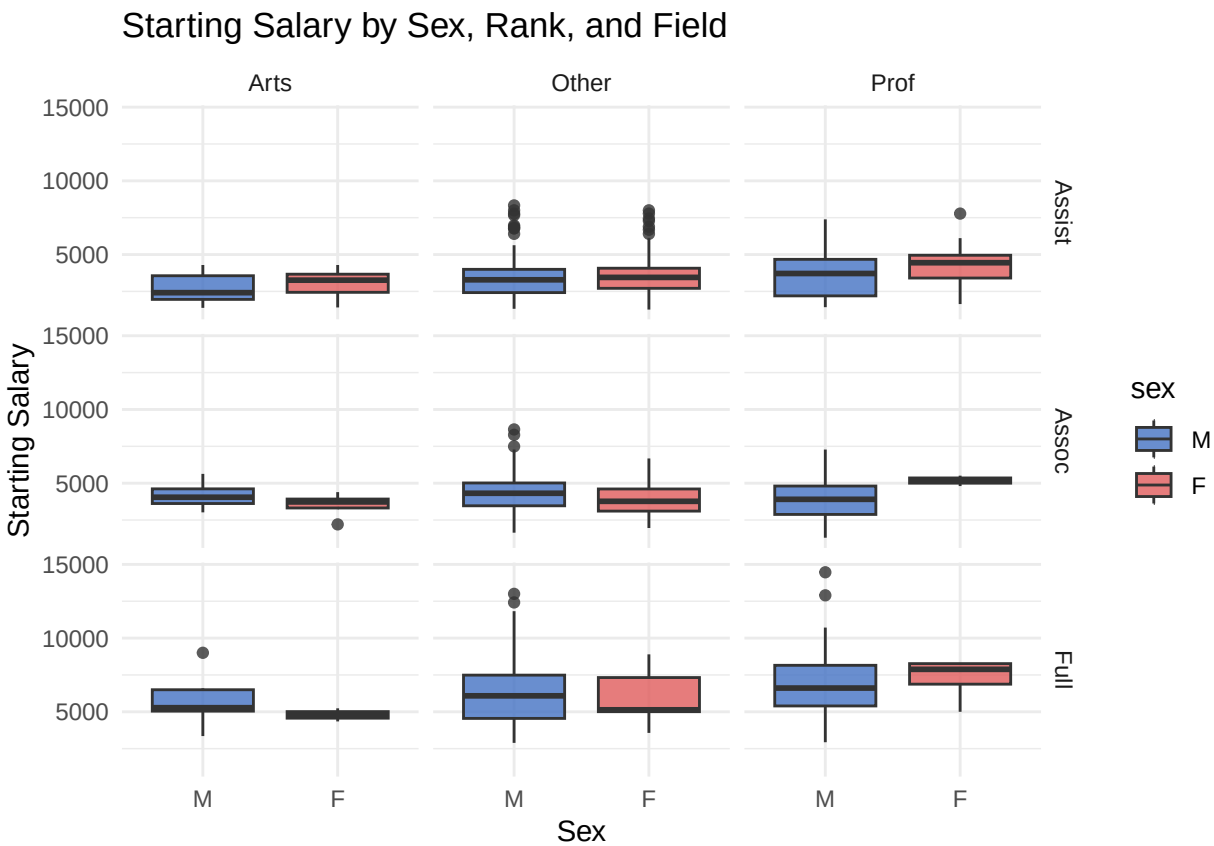

**Starting Salary by Academic Field and Sex**

The boxplots show that Professional fields exhibit the highest median starting salaries, followed by Other fields and then Arts fields. This suggests that academic field may play an important role in determining initial compensation. In both the Arts and Professional fields, the median starting salary for women are slightly higher than that for men, while the male median starting salary is slightly higher in the Other field. These patterns indicate that differences in starting salaries may vary across academic fields, highlighting the importance for controlling for field.




**Starting Salary by Sex, Rank, and Field**

When starting salaries are examined jointly by rank and field, the median salary is not consistently higher for one gender across all groups. Notably, within the Professional field, women have higher median starting salaries than men at all three ranks (Assistant, Associate, and Full). In the Arts field and Other field, male median starting salaries are higher at Associate and Full ranks, while female median starting salary is higher at the Assistant rank. Therefore, both academic rank and field should be controlled for in subsequent analyses.




**Naive Statistical Test**

We test the null hypothesis $H_0 : \mu_M - \mu_F = 0$ against the two-sided alternative $H_A : \mu_M - \mu_F \neq 0$, where $\mu_M$ and $\mu_F$ denote the population mean starting salaries for male and female faculty, respectively.

We conducted a Welch two-sample t-test to determine whether the difference in means is statistically significant. Since the histograms of starting salary distribution by sex show deviation from normality, we also perform a z-test based on the normal approximation to compare mean starting salaries between men and women. In both tests, we obtain a test statistic of 4.1326 and a p-value less than 0.001. The observed difference in starting salaries between men and women is 4.13 standard errors away from zero under the null hypothesis. The small p-value provides strong statistical evidence against the null.

However, this comparison is naive because it does not account for other factors that influence starting salaries, such as academic field and rank. To better isolate the relationship between gender and starting salary, we next fit a regression model that control for these relevant characteristics.

**Regression Analysis**

To further investigate whether gender differences in starting salaries persist after accounting for other relevant characteristics, we estimate a linear regression model of the form:

$salary_i = \beta_0 + \beta_1\, sexF_i + \beta_2\, I(rank_i = \text{Assoc}) + \beta_3\, I(rank_i = \text{Full}) + \beta_4\, I(field_i = \text{Other}) + \beta_5\, I(field_i = \text{Prof}) + \beta_6\, yrdeg_i + \beta_7\, startyr_i + \epsilon_i$

All predictors in the model are statistically significant at the 5% level. The regression results show that the coefficient for sexF is -197.94, indicating that, holding rank, field, degree year, and hire year constant, women earn on average approximately \$198 less in starting salary per month than men. The corresponding p-value is 0.00199. On average and controlling for other predictors, faculty hired directly as Associate Professors earn approximately \$686 more per month than Assistant Professors, while those hired as Full Professors

earn around $2624 more per month than Assistant Professors. Adjusting for other variables, starting salaries in Other fields exceed those in Arts by roughly $397 on average, and Professional fields exceed Arts on average by approximately $920. Starting salaries increase substantially over time, as reflected by the positive coefficient on hiring year. In addition, faculty who obtained their highest degree more recently tend to receive slightly lower starting salaries, holding other variables constant.

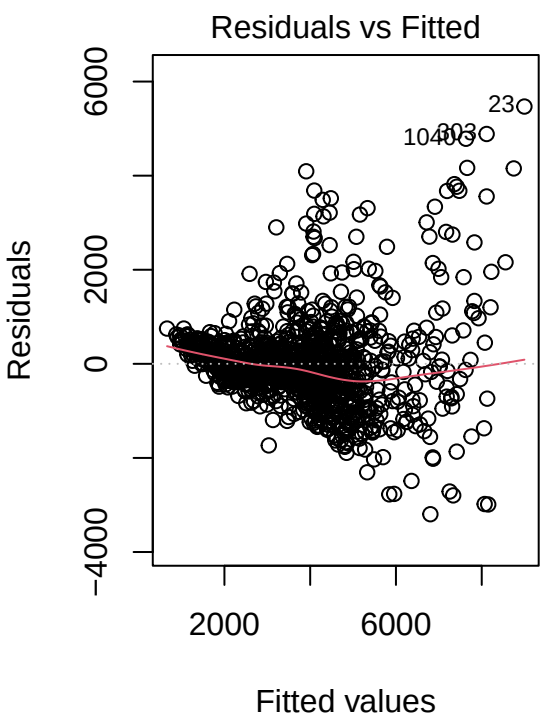
Overall, the model explains a substantial portion of the variation in starting salaries, with an R-squared value of 0.72. These results suggest that rank, field, and hiring year are major determinants of starting salary, and that a statistically significant gender difference remains even after controlling for these factors.

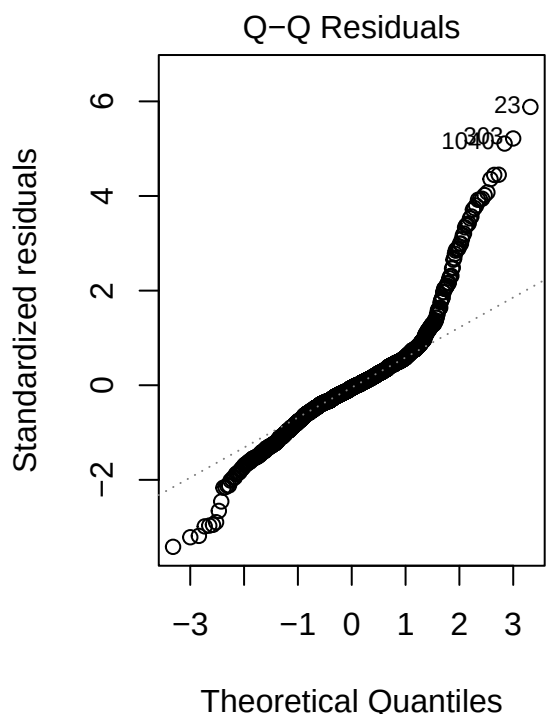
```
##                 Estimate Std. Error    t value      Pr(>|t|)
## (Intercept) -13401.32174 445.760433 -30.063955 2.123673e-145
## sexF          -197.93971  63.872183  -3.098997  1.991105e-03
## rankAssoc      685.66505  88.268504   7.767947  1.821433e-14
## rankFull      2624.01827 125.787193  20.860775  1.107675e-81
## fieldOther     396.96589  86.176200   4.606445  4.574801e-06
## fieldProf      919.77714 101.849528   9.030745  7.472752e-19
## yrdeg          -27.53572   6.047770  -4.553037  5.877356e-06
## startyr        215.55063   7.241792  29.764818 2.961717e-143
```

```
##
## R-squared: 0.7210549 ,  Adjusted R-squared: 0.7192782
```

**Regression Diagnostics**

The Residuals vs Fitted plot shows increasing spread in the residuals as fitted values increase, indicating heteroskedasticity and suggesting that the variance of errors grows with salary. The Q–Q plot shows noticeable deviations from the reference line in the upper tail, indicating that the residuals are not normally distributed, likely due to the right-skewed salary distribution. To account for the potential heteroskedasticity, we also estimated the model using heteroskedasticity-robust standard errors, which yielded the same conclusions regarding statistical significance for all predictors. Because salaries are positively skewed and variability increases with salary level, modeling log(salary) helps stabilize the variance and produce residuals that better satisfy the regression assumptions.



**Log-Salary Model with Robust Standard Errors**

The log-salary regression was re-estimated using heteroskedasticity-robust standard errors to account for potential non-constant variance in the error terms. Out of all predictors, yrdeg (year highest degree attained) was the only variable that was no longer statistically significant, while all other coefficients, including the gender coefficient, remained statistically significant. The log of salary for women is 0.0498 lower than for men, controlling for other variables. After exponentiating, women earn about 4.85% lower starting salaries than men. Overall, our Question 1 analysis suggests that women receive lower starting salaries than men at the university. This difference persists even after controlling for rank, academic field, and experience-related factors in the regression models.

```
##
## t test of coefficients:
##
##               Estimate Std. Error t value  Pr(>|t|)
## (Intercept)  3.1455228  0.0845574 37.1999 < 2.2e-16 ***
## sexF        -0.0497548  0.0119122 -4.1768 3.191e-05 ***
## rankAssoc    0.2286783  0.0182214 12.5500 < 2.2e-16 ***
## rankFull     0.5877590  0.0264936 22.1849 < 2.2e-16 ***
## fieldOther   0.1070116  0.0108100  9.8993 < 2.2e-16 ***
## fieldProf    0.2283944  0.0165024 13.8401 < 2.2e-16 ***
## yrdeg       -0.0022370  0.0013591 -1.6459    0.1001
## startyr      0.0576450  0.0015716 36.6798 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Approximate percent difference for women in starting salary:
```

```
## -4.85 %
```

## 4a. Q2: Wage difference at promotion

```r
promotion_data <- salary_clean %>%
  arrange(id, year) %>%
  group_by(id) %>%
  mutate(
    prev_rank = lag(rank),
    prev_salary = lag(salary),
    prev_year = lag(year)
  ) %>%
  ungroup() %>%
  filter(prev_rank == "Assoc", rank == "Full")

promotion_data %>%
  summarise(
    n_promotions = n(),
    n_people = n_distinct(id)
  ) %>%
  print()
```

```
## # A tibble: 1 x 2
##   n_promotions n_people
##          <int>    <int>
## 1          547      547
```

```r
promotion_data <- promotion_data %>%
  group_by(id) %>%
  slice(1) %>%
  ungroup()
```

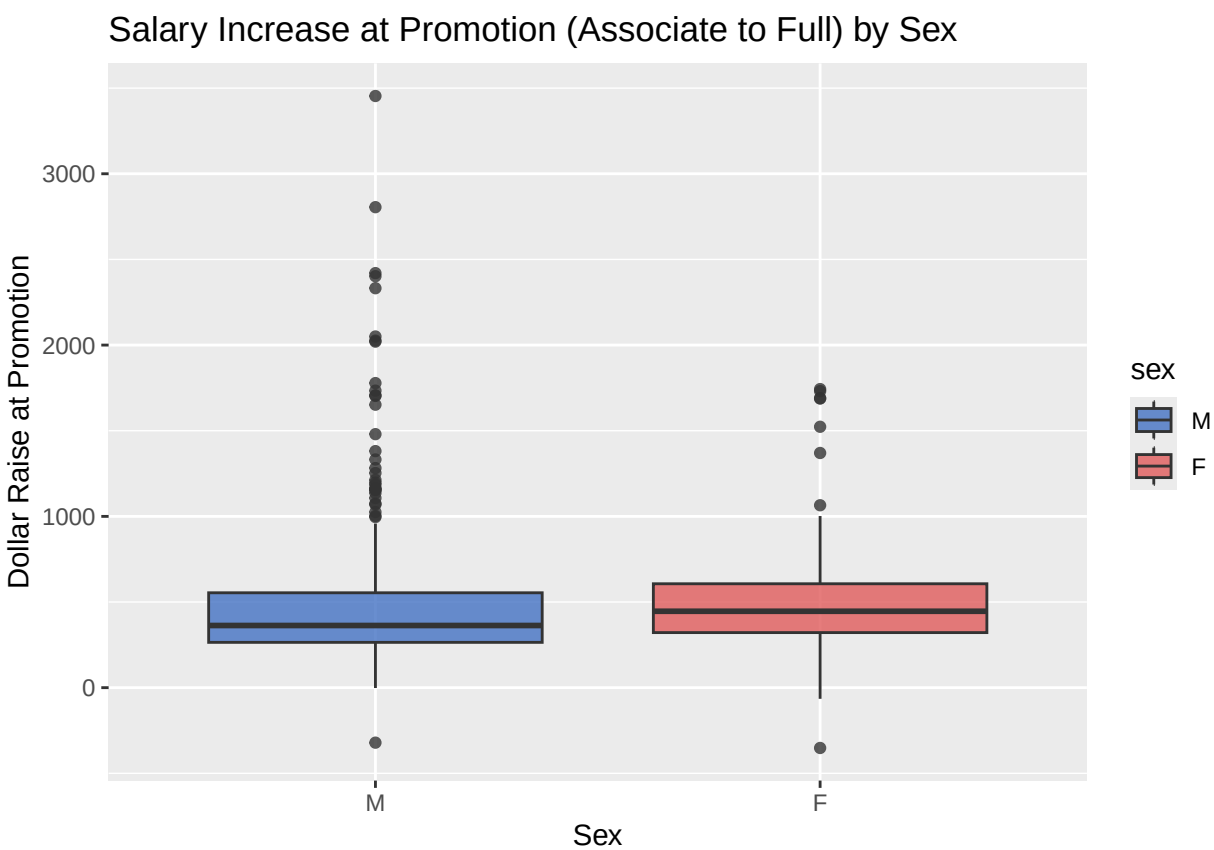
```r
promotion_data <- promotion_data %>%
  mutate(
    raise_amount = salary - prev_salary,
    raise_pct = (salary - prev_salary) / prev_salary
  )

cat("\nPromotion raise summaries by sex:\n")
```

```
##
## Promotion raise summaries by sex:
```

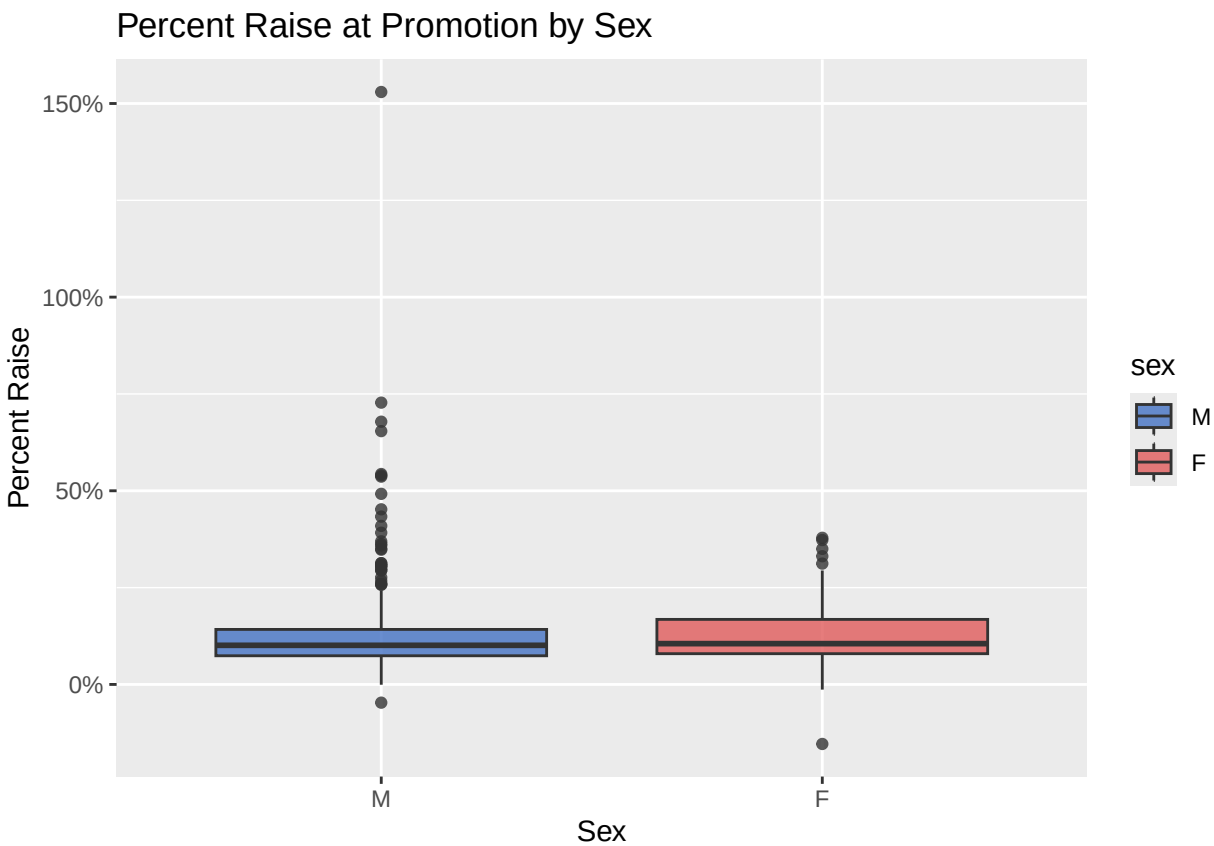
```r
promotion_data %>%
  group_by(sex) %>%
  summarise(
    n = n(),
    mean_raise = mean(raise_amount, na.rm = TRUE),
    median_raise = median(raise_amount, na.rm = TRUE),
    mean_pct_raise = mean(raise_pct, na.rm = TRUE),
    median_pct_raise = median(raise_pct, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  print()
```

```
## # A tibble: 2 x 6
##   sex       n mean_raise median_raise mean_pct_raise median_pct_raise
##   <fct> <int>      <dbl>        <dbl>          <dbl>            <dbl>
## 1 M       447       470.          363          0.124            0.101
## 2 F       100       517.          446          0.129            0.105
```

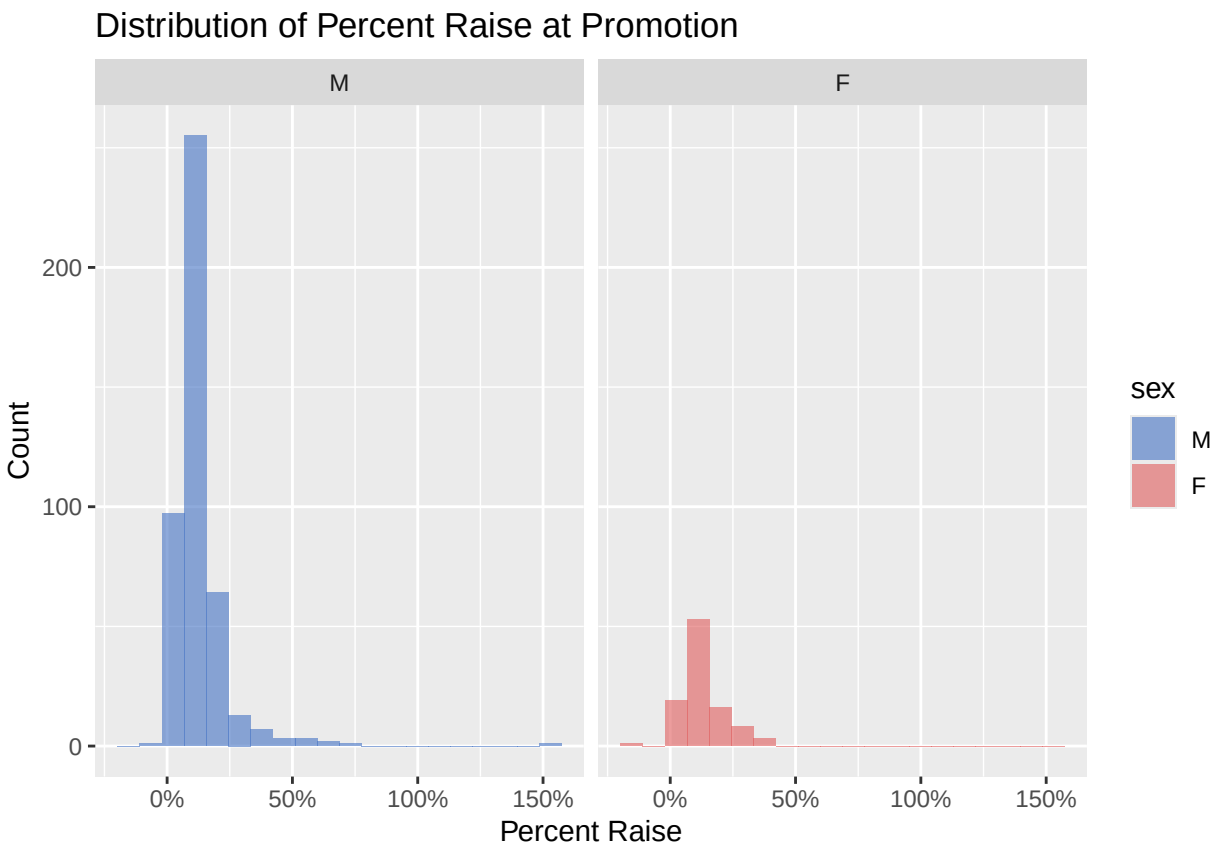
```r
ggplot(promotion_data, aes(x = sex, y = raise_amount, fill = sex)) +
  geom_boxplot(alpha = 0.8) +
  scale_fill_manual(values = sex_colours) +
  labs(
    title = "Salary Increase at Promotion (Associate to Full) by Sex",
    x = "Sex",
    y = "Dollar Raise at Promotion"
  )
```




```r
ggplot(promotion_data, aes(x = sex, y = raise_pct, fill = sex)) +
  geom_boxplot(alpha = 0.8) +
  scale_fill_manual(values = sex_colours) +
  scale_y_continuous(labels = percent_format(accuracy = 1)) +
  labs(
    title = "Percent Raise at Promotion by Sex",
    x = "Sex",
    y = "Percent Raise"
  )
```




```r
ggplot(promotion_data, aes(x = raise_pct, fill = sex)) +
  geom_histogram(alpha = 0.6, bins = 20, position = "identity") +
  scale_fill_manual(values = sex_colours) +
  facet_wrap(~ sex) +
  scale_x_continuous(labels = percent_format(accuracy = 1)) +
  labs(
    title = "Distribution of Percent Raise at Promotion",
    x = "Percent Raise",
    y = "Count"
  )
```




```r
cat("\nQ2 Welch t-test for dollar raise:\n")
```

```
##
## Q2 Welch t-test for dollar raise:
```

```r
print(t.test(raise_amount ~ sex, data = promotion_data))
```

```
##
##  Welch Two Sample t-test
##
## data:  raise_amount by sex
## t = -1.1385, df = 155.45, p-value = 0.2566
## alternative hypothesis: true difference in means between group M and group F is not equal to 0
## 95 percent confidence interval:
##  -127.73859   34.32727
## sample estimates:
## mean in group M mean in group F
##        469.9813        516.6870
```

```r
cat("\nQ2 Welch t-test for percent raise:\n")
```

```
##
## Q2 Welch t-test for percent raise:
```

```r
print(t.test(raise_pct ~ sex, data = promotion_data))
```

```
##
##  Welch Two Sample t-test
##
## data:  raise_pct by sex
```

```
## t = -0.45167, df = 182.87, p-value = 0.652
## alternative hypothesis: true difference in means between group M and group F is not equal to 0
## 95 percent confidence interval:
##  -0.02508576  0.01573983
## sample estimates:
## mean in group M mean in group F
##        0.124312        0.128985
```

```r
cat("\nQ2 Wilcoxon test for percent raise:\n")
```

```
##
## Q2 Wilcoxon test for percent raise:
```

```r
print(wilcox.test(raise_pct ~ sex, data = promotion_data))
```

```
##
##  Wilcoxon rank sum test with continuity correction
##
## data:  raise_pct by sex
## W = 20318, p-value = 0.155
## alternative hypothesis: true location shift is not equal to 0
```

```r
q2_lm_raise <- lm(
  raise_amount ~ sex + field + yrdeg + prev_salary,
  data = promotion_data
)
cat("\nQ2 regression for dollar raise:\n")
```

```
##
## Q2 regression for dollar raise:
```

```r
summary(q2_lm_raise)
```

```
##
## Call:
## lm(formula = raise_amount ~ sex + field + yrdeg + prev_salary,
##     data = promotion_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1052.51  -191.06   -77.61    83.92  3059.16
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -6.858e+02  2.296e+02  -2.987  0.00295 **
## sexF         2.586e+01  4.282e+01   0.604  0.54612
## fieldOther  -8.941e+01  4.768e+01  -1.875  0.06128 .
## fieldProf    1.250e+00  5.730e+01   0.022  0.98261
## yrdeg        1.675e+01  3.629e+00   4.616 4.89e-06 ***
## prev_salary -1.156e-03  1.574e-02  -0.073  0.94150
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 377.3 on 541 degrees of freedom
## Multiple R-squared:  0.07412,    Adjusted R-squared:  0.06556
## F-statistic: 8.661 on 5 and 541 DF,  p-value: 6.54e-08
```

```r
cat("\nQ2 regression for dollar raise with robust SE:\n")
```

```
##
## Q2 regression for dollar raise with robust SE:
```

```r
coeftest(q2_lm_raise, vcov = vcovHC(q2_lm_raise, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                Estimate  Std. Error t value  Pr(>|t|)
## (Intercept) -685.789742  233.807443 -2.9331  0.003498 **
## sexF          25.860075   40.355920  0.6408  0.521924
## fieldOther   -89.409015   48.342644 -1.8495  0.064933 .
## fieldProf      1.249621   61.724909  0.0202  0.983855
## yrdeg         16.752126    3.920405  4.2731 2.278e-05 ***
## prev_salary   -0.001156    0.015454 -0.0748  0.940400
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
q2_lm_pct <- lm(
  raise_pct ~ sex + field + yrdeg + prev_salary,
  data = promotion_data
)
cat("\nQ2 regression for percent raise:\n")
```

```
##
## Q2 regression for percent raise:
```

```r
summary(q2_lm_pct)
```

```
##
## Call:
## lm(formula = raise_pct ~ sex + field + yrdeg + prev_salary, data = promotion_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.27621 -0.05158 -0.02007  0.02561  1.37505
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -4.893e-02  6.478e-02  -0.755   0.4504
## sexF         2.302e-03  1.208e-02   0.191   0.8489
## fieldOther  -2.593e-02  1.345e-02  -1.928   0.0543 .
## fieldProf   -1.316e-03  1.616e-02  -0.081   0.9351
## yrdeg        4.190e-03  1.024e-03   4.093 4.90e-05 ***
## prev_salary -2.828e-05  4.442e-06  -6.366 4.13e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1064 on 541 degrees of freedom
## Multiple R-squared:  0.07925,    Adjusted R-squared:  0.07074
## F-statistic: 9.313 on 5 and 541 DF,  p-value: 1.6e-08
```

```r
cat("\nQ2 regression for percent raise with robust SE:\n")
```

```
##
```

```
## Q2 regression for percent raise with robust SE:
```

```r
coeftest(q2_lm_pct, vcov = vcovHC(q2_lm_pct, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                Estimate  Std. Error t value  Pr(>|t|)
## (Intercept) -4.8927e-02  5.7037e-02 -0.8578   0.39138
## sexF         2.3017e-03  1.0238e-02  0.2248   0.82220
## fieldOther  -2.5934e-02  1.3012e-02 -1.9930   0.04676 *
## fieldProf   -1.3165e-03  1.6098e-02 -0.0818   0.93485
## yrdeg        4.1904e-03  9.8829e-04  4.2401 2.628e-05 ***
## prev_salary -2.8276e-05  4.7286e-06 -5.9799 4.059e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

### 4b. Q2: Promotion Rates and Within-Rank Wage Gaps

This section investigates whether there are gender differences in promotion rates from Associate Professor to Full Professor, as well as differences in salary within academic ranks. Specifically, we examine (1) whether men and women are promoted to Full Professor at different rates, (2) how long faculty spend as Associate Professors before promotion, and (3) whether women earn systematically different salaries within the Associate and Full ranks after controlling for relevant characteristics.

**Method** We first construct a **person-level dataset** restricted to faculty who have ever held the rank of Associate Professor. For each individual, we identify whether they were eventually promoted to Full Professor, the year they first became an Associate Professor, and the year they were first promoted to Full Professor (if applicable). We also compute the number of years spent as an Associate Professor before promotion. Additionally, we record salary at Associate entry, the last Associate salary, and the salary at entry into the Full Professor rank.

This person-level dataset allows us to compare promotion outcomes and salary changes across individuals.

```r
# build person-level dataset restricted to ever-Associate faculty
ever_assoc_ids <- salary_clean %>%
  group_by(id) %>%
  filter(any(rank == "Assoc")) %>%
  pull(id) %>%
  unique()

person_level <- salary_clean %>%
  filter(id %in% ever_assoc_ids) %>%
  arrange(id, year) %>%
  group_by(id) %>%
  summarise(
    sex             = first(sex),
    deg             = first(deg),
    field           = first(field),
    yrdeg           = first(yrdeg),
    promoted        = as.integer(any(rank == "Full")),
    yr_first_assoc  = min(year[rank == "Assoc"], na.rm = TRUE),
    yr_first_full   = ifelse(any(rank == "Full"),
                             min(year[rank == "Full"], na.rm = TRUE), NA_real_),
    yrs_as_assoc    = ifelse(any(rank == "Full"),
                             yr_first_full - yr_first_assoc, NA_real_),
```

```r
    sal_assoc_entry = salary[rank == "Assoc"][which.min(year[rank == "Assoc"])],
    sal_assoc_last  = salary[rank == "Assoc"][which.max(year[rank == "Assoc"])],
    sal_full_entry  = ifelse(any(rank == "Full"),
                             salary[rank == "Full"][which.min(year[rank == "Full"])],
                             NA_real_),
    .groups = "drop"
  )
```

**Promotion Rates by Sex** We first compare promotion rates between men and women by computing the proportion of Associate Professors who were eventually promoted to Full Professor by the end of the observation period.

```r
cat("Promotion rates by sex:\n")
```

```
## Promotion rates by sex:
```

```r
person_level %>%
  group_by(sex) %>%
  summarise(
    n          = n(),
    n_promoted = sum(promoted),
    pct        = round(100 * mean(promoted), 1),
    .groups = "drop"
  ) %>%
  print()
```

```
## # A tibble: 2 x 4
##   sex       n n_promoted   pct
##   <fct> <int>      <int> <dbl>
## 1 M       746        447  59.9
## 2 F       238        100  42
```

```r
ct <- table(Sex = person_level$sex, Promoted = person_level$promoted)
colnames(ct) <- c("Not Promoted", "Promoted")
cat("\nContingency table:\n")
```

```
##
## Contingency table:
```

```r
print(ct)
```

```
##    Promoted
## Sex Not Promoted Promoted
##   M          299      447
##   F          138      100
```

```r
cat("\nRow proportions:\n")
```

```
##
## Row proportions:
```

```r
print(round(prop.table(ct, margin = 1), 3))
```

```
##    Promoted
## Sex Not Promoted Promoted
##   M        0.401    0.599
##   F        0.580    0.420
```

```r
cat("\nMedian time as Associate before promotion:\n")
```

```
##
## Median time as Associate before promotion:
```

```r
person_level %>%
  filter(promoted == 1) %>%
  group_by(sex) %>%
  summarise(median_yrs = median(yrs_as_assoc, na.rm = TRUE), .groups = "drop") %>%
  print()
```

```
## # A tibble: 2 x 2
##   sex   median_yrs
##   <fct>      <dbl>
## 1 M              5
## 2 F              6
```

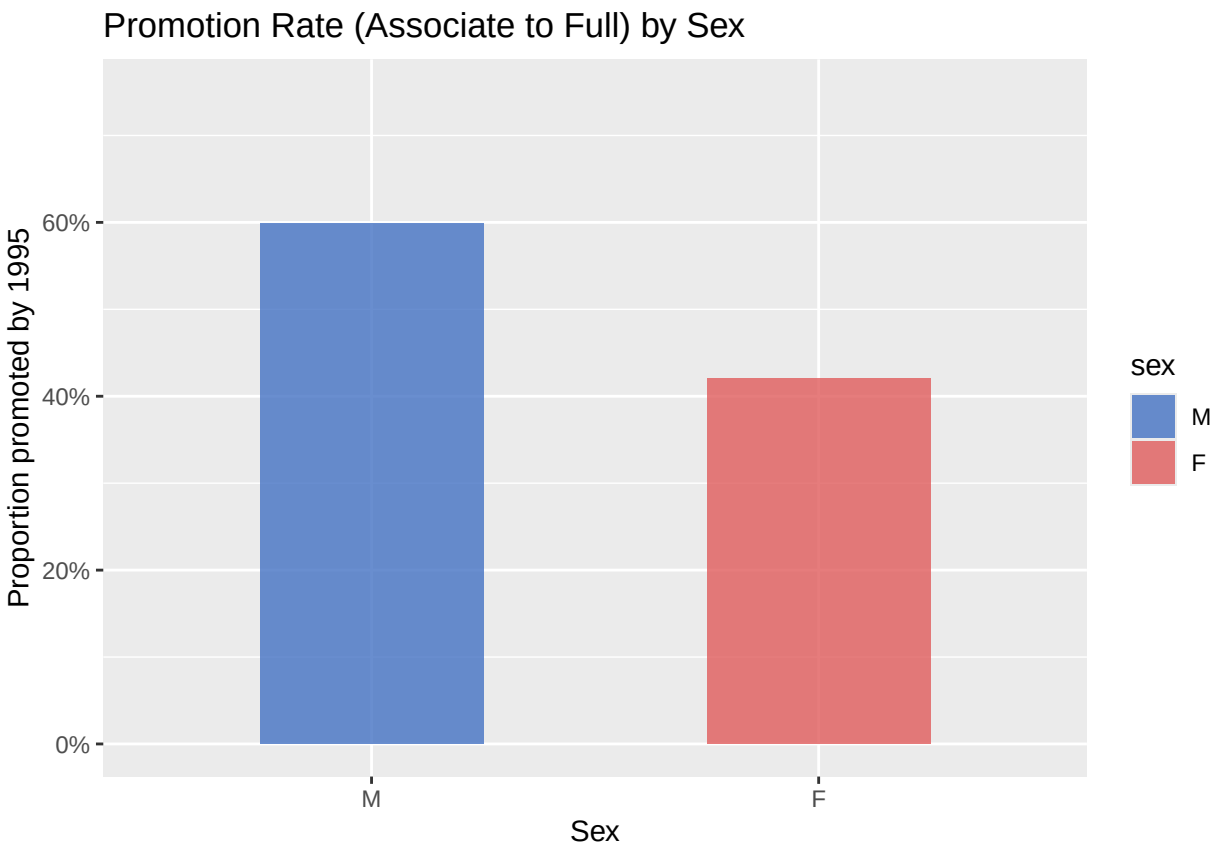
**Observations** The results show that 59.9% of male Associate Professors were promoted to Full Professor, compared to 42.0% of female Associate Professors. The contingency table indicates that 447 of 746 men were promoted, while only 100 of 238 women were promoted.

Examining row proportions shows that about 60% of men were promoted compared to 42% of women, suggesting a substantial gender gap in promotion outcomes.

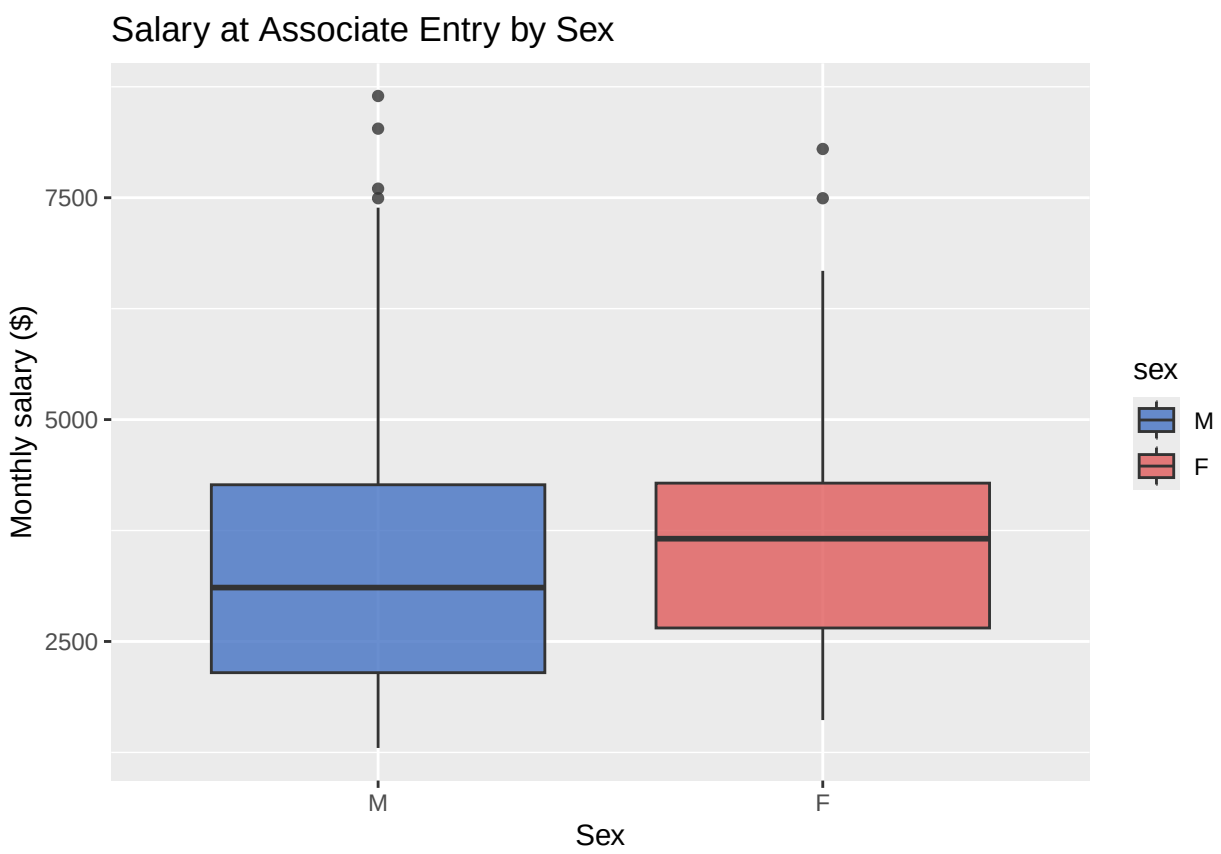
Among those who were promoted, the median time spent as an Associate Professor was 5 years for men and 6 years for women, indicating that women tend to remain in the Associate rank slightly longer before promotion.

**Visualization of Promotion and Salary Patterns** To better understand these patterns, we visualize promotion rates, entry salaries at the Associate rank, salary gains upon promotion, and the distribution of time spent as an Associate Professor.

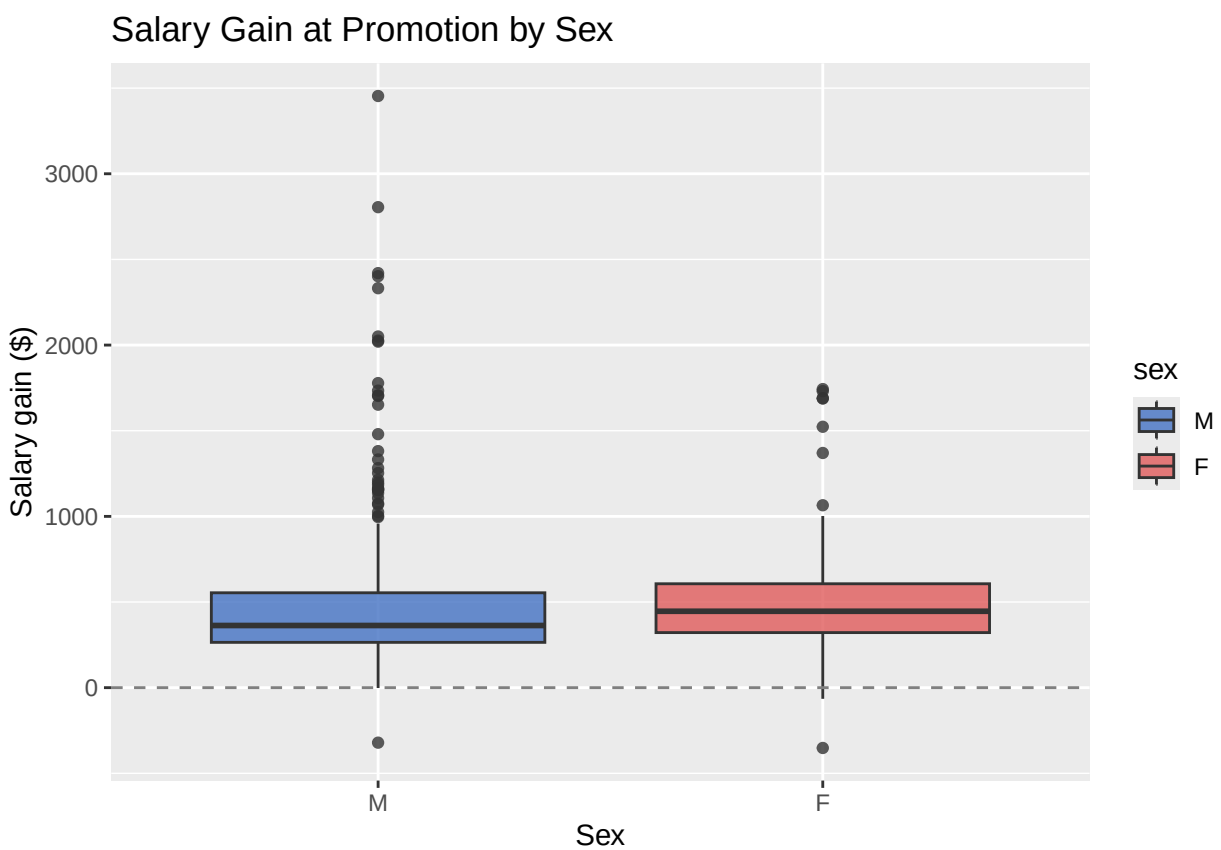
```r
# promotion rate bar chart
ggplot(person_level %>% group_by(sex) %>% summarise(pct = mean(promoted)),
       aes(x = sex, y = pct, fill = sex)) +
  geom_col(width = 0.5, alpha = 0.8) +
  scale_fill_manual(values = sex_colours) +
  scale_y_continuous(labels = percent_format(accuracy = 1), limits = c(0, 0.75)) +
  labs(title = "Promotion Rate (Associate to Full) by Sex",
       x = "Sex", y = "Proportion promoted by 1995")
```



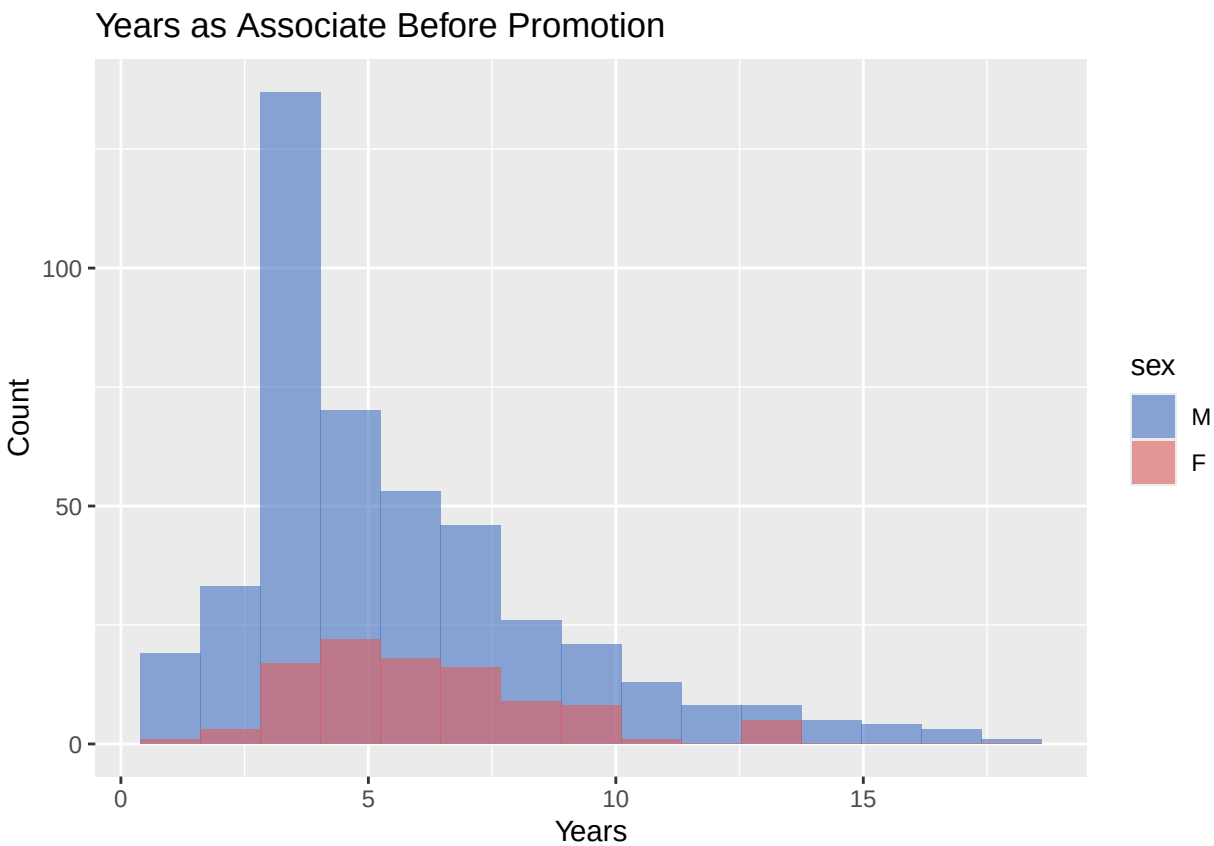

```r
# salary at Associate entry by sex
ggplot(person_level, aes(x = sex, y = sal_assoc_entry, fill = sex)) +
  geom_boxplot(alpha = 0.8) +
  scale_fill_manual(values = sex_colours) +
  labs(title = "Salary at Associate Entry by Sex",
       x = "Sex", y = "Monthly salary ($)")
```




```r
# salary gain at promotion by sex
ggplot(person_level %>% filter(promoted == 1) %>%
         mutate(sal_gain = sal_full_entry - sal_assoc_last),
       aes(x = sex, y = sal_gain, fill = sex)) +
  geom_boxplot(alpha = 0.8) +
  scale_fill_manual(values = sex_colours) +
  geom_hline(yintercept = 0, linetype = "dashed", colour = "grey50") +
  labs(title = "Salary Gain at Promotion by Sex",
       x = "Sex", y = "Salary gain ($)")
```




```r
# time as Associate before promotion
ggplot(person_level %>% filter(promoted == 1),
       aes(x = yrs_as_assoc, fill = sex)) +
  geom_histogram(alpha = 0.6, bins = 15, position = "identity") +
  scale_fill_manual(values = sex_colours) +
  labs(title = "Years as Associate Before Promotion",
       x = "Years", y = "Count")
```




#### Statistical Test for Differences in Promotion Rates To formally test whether the difference in promotion rates between men and women is statistically significant, we conduct a two-sample z-test for the difference in proportions. This test evaluates the null hypothesis that the promotion probability is the same for men and women.

```
# two-sample z-test for difference in promotion proportions
nF <- sum(person_level$sex == "F")
nM <- sum(person_level$sex == "M")
kF <- sum(person_level$sex == "F" & person_level$promoted == 1)
kM <- sum(person_level$sex == "M" & person_level$promoted == 1)
pF <- kF / nF
pM <- kM / nM

p_pool <- (kF + kM) / (nF + nM)
se_z   <- sqrt(p_pool * (1 - p_pool) * (1/nF + 1/nM))
z_stat <- (pF - pM) / se_z
p_val  <- 2 * pnorm(-abs(z_stat))

se_diff <- sqrt(pF*(1-pF)/nF + pM*(1-pM)/nM)
ci_lo   <- (pF - pM) - 1.96 * se_diff
ci_hi   <- (pF - pM) + 1.96 * se_diff

cat(sprintf("pF = %.4f  (%d/%d)\n", pF, kF, nF))
```

```
## pF = 0.4202  (100/238)
```

```
cat(sprintf("pM = %.4f  (%d/%d)\n", pM, kM, nM))
```

```
## pM = 0.5992  (447/746)
```

```r
cat(sprintf("Difference pF - pM = %+.4f\n", pF - pM))
```

```
## Difference pF - pM = -0.1790
```

```r
cat(sprintf("95%% CI: (%.4f, %.4f)\n", ci_lo, ci_hi))
```

```
## 95% CI: (-0.2509, -0.1071)
```

```r
cat(sprintf("z = %.3f,  p = %.4f\n", z_stat, p_val))
```

```
## z = -4.840,  p = 0.0000
```

```r
cat("\nEquivalent chi-square test:\n")
```

```
##
## Equivalent chi-square test:
```

```r
print(chisq.test(ct, correct = FALSE))
```

```
##
##  Pearson's Chi-squared test
##
## data:  ct
## X-squared = 23.425, df = 1, p-value = 1.299e-06
```

**Observations** The promotion rate for women is 0.420, while the promotion rate for men is 0.599, producing a difference of -0.179. The 95% confidence interval for this difference is (-0.251, -0.107), which does not include zero.

The z-test statistic is -4.84, with a p-value less than 0.001, indicating that the difference in promotion rates between men and women is statistically significant. The equivalent chi-square test yields the same conclusion.

These results provide strong evidence that women were less likely to be promoted from Associate to Full Professor during the sample period.

**Within-Rank Wage Gap Analysis** Next, we examine whether salary differences persist within academic ranks after controlling for relevant characteristics. For each rank (Associate and Full Professor), we estimate a log-linear regression model where the dependent variable is the logarithm of salary.

The model controls for:

sex

year of degree

calendar year

administrative roles

years of experience

academic field

degree type

Using log salary allows the coefficient on the gender indicator to be interpreted approximately as a percentage difference in salary.

```r
# log-salary regression within rank to estimate adjusted wage gap by sex
for (rnk in c("Assoc", "Full")) {
  sub <- salary_clean %>%
    filter(rank == rnk, salary > 0) %>%
```

```r
    mutate(log_salary = log(salary), years_exp = year - startyr)

  fit <- lm(
    log_salary ~ sex + yrdeg + year + admin + years_exp + field + deg,
    data = sub
  )

  cat(sprintf("\nQ2 log-salary regression - %s rank (n = %d):\n", rnk, nrow(sub)))
  summary(fit)

  cat("\nWith robust SE:\n")
  print(coeftest(fit, vcov = vcovHC(fit, type = "HC1")))

  coef_f <- coef(fit)["sexF"]
  cat(sprintf("Approximate percent difference for women in salary: %.1f%%\n",
              (exp(coef_f) - 1) * 100))
}
```

```
##
## Q2 log-salary regression - Assoc rank (n = 6529):
##
## With robust SE:
##
## t test of coefficients:
##
##                Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)  3.5985e+00  3.0646e-02 117.4231 < 2.2e-16 ***
## sexF        -5.4535e-02  4.3107e-03 -12.6512 < 2.2e-16 ***
## yrdeg        9.1606e-06  5.3397e-04   0.0172 0.9863128
## year         5.2330e-02  5.2202e-04 100.2454 < 2.2e-16 ***
## admin1       1.7104e-01  8.1262e-03  21.0481 < 2.2e-16 ***
## years_exp   -2.7630e-03  4.6623e-04  -5.9263 3.257e-09 ***
## fieldOther   8.4987e-02  4.1555e-03  20.4518 < 2.2e-16 ***
## fieldProf    2.0215e-01  5.9906e-03  33.7439 < 2.2e-16 ***
## degPhD       1.8608e-02  5.2389e-03   3.5518 0.0003853 ***
## degProf      1.4213e-01  9.2447e-03  15.3747 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate percent difference for women in salary: -5.3%
##
## Q2 log-salary regression - Full rank (n = 9211):
##
## With robust SE:
##
## t test of coefficients:
##
##                Estimate  Std. Error t value  Pr(>|t|)
## (Intercept)  3.97123426  0.03450479 115.092 < 2.2e-16 ***
## sexF        -0.05705937  0.00543111 -10.506 < 2.2e-16 ***
## yrdeg       -0.01272580  0.00044547 -28.567 < 2.2e-16 ***
## year         0.06040065  0.00050395 119.854 < 2.2e-16 ***
## admin1       0.13309786  0.00494351  26.924 < 2.2e-16 ***
## years_exp   -0.00532371  0.00038306 -13.898 < 2.2e-16 ***
```

```
## fieldOther   0.12164143  0.00568107  21.412 < 2.2e-16 ***
## fieldProf    0.22141926  0.00673904  32.856 < 2.2e-16 ***
## degPhD       0.11049705  0.00840394  13.148 < 2.2e-16 ***
## degProf      0.21449233  0.01061574  20.205 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate percent difference for women in salary: -5.5%
```

**Results** The regression results indicate a statistically significant negative coefficient on the female indicator in both ranks.

Among Associate Professors, the coefficient on sexF corresponds to an approximate salary difference of -5.3%, meaning that women earn about 5.3% less than men on average after controlling for experience, year effects, field, degree type, and administrative roles.

Among Full Professors, the estimated difference is -5.5%, indicating that women continue to earn roughly 5–6% less than men even after reaching the highest academic rank and after accounting for observable characteristics.

**Interpretation** Taken together, these results suggest two important patterns. First, women were less likely to be promoted from Associate to Full Professor during the study period. Second, conditional on rank and other observed characteristics, women earned modestly lower salaries than men within both the Associate and Full Professor ranks. These findings suggest that gender disparities may arise both through promotion dynamics and within-rank wage differences.

## 5. Q3: Overall salary discrimination

```r
# avoid longitudinal methods by using one row per faculty (latest salary)
latest_salary <- salary_clean %>%
  arrange(id, year) %>%
  group_by(id) %>%
  slice_tail(n = 1) %>%
  ungroup() %>%
  mutate(
    log_salary = log(salary),
    experience = year - startyr
  )

latest_salary %>%
  summarise(
    n_people = n(),
    min_year = min(year, na.rm = TRUE),
    max_year = max(year, na.rm = TRUE)
  ) %>%
  print()
```

```
## # A tibble: 1 x 3
##   n_people min_year max_year
##      <int>    <int>    <int>
## 1     1597       95       95
```

```r
cat("\nLatest salary summaries by sex and rank:\n")
```

```
##
```

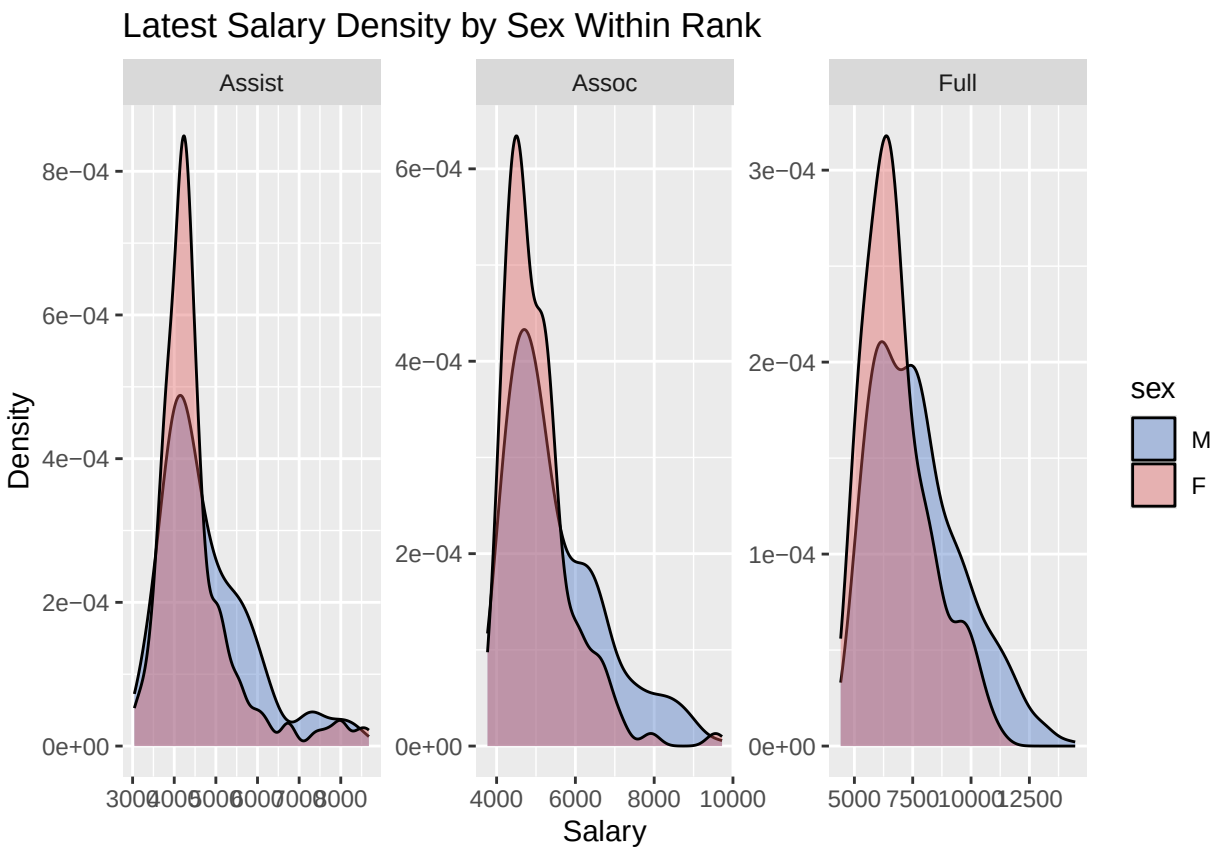
```
## Latest salary summaries by sex and rank:
```

```r
latest_salary %>%
  group_by(rank, sex) %>%
  summarise(
    n = n(),
    mean_salary = mean(salary, na.rm = TRUE),
    median_salary = median(salary, na.rm = TRUE),
    sd_salary = sd(salary, na.rm = TRUE),
    .groups = "drop"
  )
```

```
## # A tibble: 6 x 6
##   rank   sex       n mean_salary median_salary sd_salary
##   <fct>  <fct> <int>       <dbl>         <dbl>     <dbl>
## 1 Assist M       170       4774.         4430.     1117.
## 2 Assist F       145       4503.         4286       981.
## 3 Assoc  M       299       5480.         5095      1224.
## 4 Assoc  F       138       5019.         4743       858.
## 5 Full   M       719       7715.         7388      1943.
## 6 Full   F       126       6840.         6590      1436.
```

```r
ggplot(latest_salary, aes(x = sex, y = salary, fill = sex)) +
  geom_boxplot(alpha = 0.8) +
  scale_fill_manual(values = sex_colours) +
  facet_wrap(~ rank, scales = "free_y") +
  labs(
    title = "Latest Salary by Sex Within Rank",
    x = "Sex",
    y = "Salary"
  )
```

```
ggplot(latest_salary, aes(x = salary, fill = sex)) +
  geom_density(alpha = 0.4) +
  scale_fill_manual(values = sex_colours) +
  facet_wrap(~ rank, scales = "free") +
  labs(
    title = "Latest Salary Density by Sex Within Rank",
    x = "Salary",
    y = "Density"
  )
```




```r
q3_lm <- lm(
  salary ~ sex + rank + field + experience + admin,
  data = latest_salary
)
cat("\nQ3 overall salary model (latest salary):\n")
```

```
##
## Q3 overall salary model (latest salary):
```

```r
summary(q3_lm)
```

```
##
## Call:
## lm(formula = salary ~ sex + rank + field + experience + admin,
##     data = latest_salary)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3334.3  -951.5  -164.3   600.3  6702.6
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 4016.626    131.041  30.652  < 2e-16 ***
## sexF        -409.325     87.059  -4.702 2.80e-06 ***
## rankAssoc    600.365    113.938   5.269 1.56e-07 ***
## rankFull    2607.643    122.202  21.339  < 2e-16 ***
## fieldOther   781.297    106.729   7.320 3.91e-13 ***
## fieldProf   1769.944    128.711  13.751  < 2e-16 ***
```

```
## experience    -4.236      4.603  -0.920    0.358
## admin1      1202.224    118.822  10.118  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1431 on 1589 degrees of freedom
## Multiple R-squared:  0.5084, Adjusted R-squared:  0.5062
## F-statistic: 234.7 on 7 and 1589 DF,  p-value: < 2.2e-16
```

```r
cat("\nQ3 overall salary model with robust SE:\n")
```

```
##
## Q3 overall salary model with robust SE:
```

```r
coeftest(q3_lm, vcov = vcovHC(q3_lm, type = "HC1"))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value  Pr(>|t|)
## (Intercept) 4016.6261    88.3316 45.4721 < 2.2e-16 ***
## sexF        -409.3248    70.0167 -5.8461 6.095e-09 ***
## rankAssoc    600.3651    89.1285  6.7359 2.269e-11 ***
## rankFull    2607.6426   126.5920 20.5988 < 2.2e-16 ***
## fieldOther   781.2971    74.7331 10.4545 < 2.2e-16 ***
## fieldProf   1769.9444   108.0356 16.3830 < 2.2e-16 ***
## experience    -4.2364     5.6385 -0.7513    0.4526
## admin1      1202.2236   131.4548  9.1455 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# interaction model: does the salary gap differ by rank?
q3_interaction <- lm(
  salary ~ sex * rank + field + experience + admin,
  data = latest_salary
)
cat("\nQ3 salary model with sex x rank interaction:\n")
```

```
##
## Q3 salary model with sex x rank interaction:
```

```r
summary(q3_interaction)
```

```
##
## Call:
## lm(formula = salary ~ sex * rank + field + experience + admin,
##     data = latest_salary)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3374.8  -941.1  -187.9   594.1  6667.3
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3868.301    145.024  26.673  < 2e-16 ***
## sexF            -64.592    162.155  -0.398  0.69044
## rankAssoc       754.347    144.034   5.237 1.85e-07 ***
```

```
## rankFull       2817.331    143.439  19.641  < 2e-16 ***
## fieldOther      770.145    106.718   7.217 8.22e-13 ***
## fieldProf      1763.087    128.600  13.710  < 2e-16 ***
## experience       -4.811      4.598  -1.046  0.29565
## admin1         1205.407    118.575  10.166  < 2e-16 ***
## sexF:rankAssoc -314.264    218.627  -1.437  0.15079
## sexF:rankFull  -627.052    212.653  -2.949  0.00324 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1428 on 1587 degrees of freedom
## Multiple R-squared:  0.5111, Adjusted R-squared:  0.5083
## F-statistic: 184.3 on 9 and 1587 DF,  p-value: < 2.2e-16
```

```r
cat("\nQ3 interaction model with robust SE:\n")
```

```
##
## Q3 interaction model with robust SE:
```

```r
coeftest(q3_interaction, vcov = vcovHC(q3_interaction, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                 Estimate Std. Error t value  Pr(>|t|)
## (Intercept)    3868.3009    99.7946 38.7626 < 2.2e-16 ***
## sexF            -64.5917   111.5342 -0.5791 0.5625900
## rankAssoc       754.3475   114.2743  6.6012 5.545e-11 ***
## rankFull       2817.3313   144.8704 19.4472 < 2.2e-16 ***
## fieldOther      770.1445    74.4327 10.3469 < 2.2e-16 ***
## fieldProf      1763.0873   108.0410 16.3187 < 2.2e-16 ***
## experience       -4.8107     5.6345 -0.8538 0.3933496
## admin1         1205.4075   130.8587  9.2115 < 2.2e-16 ***
## sexF:rankAssoc -314.2637   145.4078 -2.1613 0.0308246 *
## sexF:rankFull  -627.0516   172.1399 -3.6427 0.0002785 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# log-salary
q3_lm_log <- lm(
  log_salary ~ sex + rank + field + experience + admin,
  data = latest_salary
)
cat("\nQ3 log-salary model:\n")
```

```
##
## Q3 log-salary model:
```

```r
summary(q3_lm_log)
```

```
##
## Call:
## lm(formula = log_salary ~ sex + rank + field + experience + admin,
##     data = latest_salary)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
```

```
## -0.48818 -0.14113 -0.01492  0.11271  0.68607
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  8.3226282  0.0182786 455.321  < 2e-16 ***
## sexF        -0.0590251  0.0121436  -4.861 1.29e-06 ***
## rankAssoc    0.1256876  0.0158930   7.908 4.84e-15 ***
## rankFull     0.4309180  0.0170457  25.280  < 2e-16 ***
## fieldOther   0.1200197  0.0148874   8.062 1.46e-15 ***
## fieldProf    0.2699668  0.0179535  15.037  < 2e-16 ***
## experience  -0.0003448  0.0006421  -0.537    0.591
## admin1       0.1670659  0.0165742  10.080  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1996 on 1589 degrees of freedom
## Multiple R-squared:  0.5678, Adjusted R-squared:  0.5659
## F-statistic: 298.2 on 7 and 1589 DF,  p-value: < 2.2e-16
```

```r
cat("\nQ3 log-salary model with robust SE:\n")
```

```
##
## Q3 log-salary model with robust SE:
```

```r
coeftest(q3_lm_log, vcov = vcovHC(q3_lm_log, type = "HC1"))
```

```
##
## t test of coefficients:
##
##                Estimate  Std. Error  t value  Pr(>|t|)
## (Intercept)  8.32262821  0.01424517 584.2420 < 2.2e-16 ***
## sexF        -0.05902512  0.01079013  -5.4703 5.212e-08 ***
## rankAssoc    0.12568758  0.01485859   8.4589 < 2.2e-16 ***
## rankFull     0.43091799  0.01811909  23.7825 < 2.2e-16 ***
## fieldOther   0.12001965  0.01133042  10.5927 < 2.2e-16 ***
## fieldProf    0.26996679  0.01490157  18.1167 < 2.2e-16 ***
## experience  -0.00034480  0.00071706  -0.4808    0.6307
## admin1       0.16706593  0.01594811  10.4756 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# coefs
q3_log_coef <- coef(q3_lm_log)["sexF"]
cat("\nApproximate percent difference for women in latest salary:\n")
```

```
##
## Approximate percent difference for women in latest salary:
```

```r
cat(round((exp(q3_log_coef) - 1) * 100, 2), "%\n")
```

```
## -5.73 %
```